

ZEET CAPSULE 250MG

DESCRIPTION

Clean, uniform, dry, telescopic type, hard gelatin capsules (bovine), shiny and not smudged or sticky. Printed "ZEET" & "PM." Colour: Cap- Blue & Body – Ivory.

Active Substance

Each capsule contains Mefenamic Acid 250mg

PHARMACODYNAMICS

Mechanism of Action

Mefenamic acid is a non-steroidal agent with demonstrated anti-inflammatory, analgesic, and antipyretic properties. Mefenamic acid was found to inhibit prostaglandin synthesis and to compete for binding at the prostaglandin receptor sites.

PHARMACOKINETICS

Absorption: Mefenamic acid is rapidly absorbed from the gastrointestinal tract. Following administration of a one gram oral dose to adults, peak plasma levels of 10 µg/mL occur in 1 to 4 hours, with a half-life of 2 hours. Plasma levels are proportional to dose, following multiple doses, with no drug accumulation. One gram of mefenamic acid administered four times daily produces peak blood levels of 20 µg/mL by the second day of administration.

Distribution: Mefenamic acid is extensively bound to plasma proteins.

Metabolism: Mefenamic acid metabolism is predominantly mediated via cytochrome P450 (CYP) 2C9 in the liver. Patients who are known or suspected to be poor CYP2C9 metabolizers based on previous history/experience with other CYP2C9 substrates should be administered mefenamic acid with caution as they may have abnormally high plasma levels due to reduced metabolic clearance.

Elimination: Following a single oral dose, 52% to 67% of the dose was recovered from the urine as unchanged drug or one of two metabolites. Assay of stools over 3 days accounted for 20% to 25% of the dose, chiefly as unconjugated metabolite II.

INDICATIONS

1. The symptomatic relief of rheumatic arthritis (including Still's Disease), osteoarthritis, and pain including muscular, traumatic, and dental pain, headaches of most aetiology, post-operative, and postpartum pain
2. The symptomatic relief of primary dysmenorrhea
3. Menorrhagia due to dysfunctional causes or the presence of an intrauterine device (IUD) when organic pelvic pathology has been excluded
4. Premenstrual syndrome
5. The relief of pyrexia in paediatric patients over 6 months of age

RECOMMENDED DOSAGE

Undesirable effects may be minimized by using the minimum effective dose for the shortest duration necessary to control symptoms.

The oral dosage form of mefenamic acid may be taken with food if gastrointestinal upset occurs.

Mild to moderate pain/rheumatoid arthritis/osteoarthritis in adults and adolescents over 14 years of age: 500 mg three times daily.

Dysmenorrhea: 500 mg three times daily, to be administered at the onset of menstrual pain and continued while symptoms persist according to the judgement of the physician.

Menorrhagia: 500 mg three times daily, starting with the onset of bleeding and associated symptoms and continued according to the judgement of the physician.

Premenstrual syndrome: 500 mg three times daily, starting with the onset of symptoms and continued until the anticipated cessation of symptoms according to the judgement of the physician.

For Still's Disease or antipyretic action in infants and children over 6 months to 14 years: 19.5 mg/kg to 25 mg/kg of body weight daily in divided doses three times daily.

Paediatric use

Mefenamic acid is reported to be effective for pyrexia in paediatric patients over 6 months of age and for pain in adolescents over 14 years of age.

Use in the elderly

Impairment of renal function, sometimes leading to acute renal failure, has been reported. Elderly or debilitated patients seem unable to tolerate ulceration or bleeding as well as some other individuals; most spontaneous reports of fatal gastrointestinal events are in this patient population.

ROUTE OF ADMINISTRATION: Oral

CONTRAINDICATIONS

Mefenamic acid should not be used in patients with known hypersensitivity to mefenamic acid or any of the components of this product.

Because the potential exists for cross-sensitivity to aspirin or other nonsteroidal anti-inflammatory drugs (NSAIDs), mefenamic acid should not be given to patients in whom these drugs induce symptoms of bronchospasm, allergic rhinitis or urticaria.

Mefenamic acid is contraindicated in patients with active ulceration or chronic inflammation of either the upper or lower gastrointestinal tract and should be avoided in patients with pre-existing renal disease.

Treatment of peri-operative pain in the setting of coronary artery bypass graft (CABG) surgery.

Patient with severe renal and hepatic failure.

Patients with severe heart failure.

WARNINGS & PRECAUTIONS

RISK OF GI ULCERATION, BLEEDING AND PERFORATION WITH NSAID

Serious GI toxicity such as bleeding, ulceration and perforation can occur at any time, with or without warning symptoms, in patients treated with NSAID therapy. Although minor upper GI problems (e.g. dyspepsia) are common, usually developing early in

therapy, prescribers should remain alert for ulceration and bleeding in patients treated with NSAIDs even in the absence of previous GI tract symptoms.

Studies to date have not identified any subset of patients not at risk of developing peptic ulceration and bleeding. Patients with prior history of serious GI events and other risk factors associated with peptic ulcer disease (e.g. alcoholism, smoking and corticosteroid therapy) are at increased risk. Elderly or debilitated patients seem to tolerate ulceration or bleeding less than other individuals and account for most spontaneous reports for fatal GI events.

The use of mefenamic acid with concomitant systemic non-aspirin NSAIDs including cyclooxygenase-2 (COX-2) inhibitors should be avoided. Concomitant use of a systemic NSAID and another systemic NSAID may increase frequency of gastrointestinal ulcers and bleeding.

Cardiovascular Effects: NSAIDs may cause an increased risk of serious cardiovascular (CV) thrombotic events, myocardial infarction, and stroke, which can be fatal. This risk may increase with duration of use. The relative increase of this risk appears to be similar in those with or without known CV disease or CV risk factors. However, patients with known CV disease or CV risk factors may be at greater risk in terms of absolute incidence, due to their increased rate at baseline. To minimize the potential risk for an adverse CV event in patients treated with mefenamic acid, the lowest effective dose should be used for the shortest duration possible. Physicians and patients should remain alert for the development of such events, even in the absence of previous CV symptoms. Patients should be informed about the signs and/or symptoms of serious CV toxicity and the steps to take if they occur.

Hypertension: As with all NSAIDs, mefenamic acid can lead to the onset of new hypertension or worsening the pre-existing hypertension, either of which may contribute to the increased incidence of CV events. NSAIDs, including mefenamic acid, should be used with caution in patients with hypertension. Blood pressure should be monitored closely during initiation of therapy with mefenamic acid and throughout the course of therapy.

Fluid Retention and Oedema: As with other drugs known to inhibit prostaglandin synthesis, fluid retention and oedema have been observed in some patients taking NSAIDs, including mefenamic acid. Therefore, mefenamic acid should be used with caution in patients with compromised cardiac function and other conditions predisposing to, or worsened by, fluid retention. Patients with pre-existing congestive heart failure or hypertension should be closely monitored.

Gastrointestinal (GI) Effects: If diarrhoea occurs, the dosage should be reduced or temporarily suspended. Symptoms may recur in certain patients following subsequent exposure.

NSAIDs, including mefenamic acid, can cause serious gastrointestinal (GI) adverse events including inflammation, bleeding, ulceration, and perforation of the stomach, small intestine, or large intestine, which can be fatal. When GI bleeding or ulceration occurs in patients receiving mefenamic acid, the treatment should be withdrawn. Patients most at risk of developing these

types of GI complications with NSAIDs are the elderly, patients with CV disease, patients using concomitant corticosteroids, antiplatelet drugs (such as aspirin), selective serotonin reuptake inhibitors, patients ingesting alcohol or patients with a prior history of, or active, gastrointestinal disease, such as ulceration, GI bleeding or inflammatory conditions. Therefore, mefenamic acid should be used with caution in these patients.

Skin Reactions:

Serious skin reactions, some of them are fatal, including drug reaction with eosinophilia and systemic symptoms (DRESS syndrome), exfoliative dermatitis, Stevens-Johnson Syndrome (SJS), toxic epidermal necrolysis (TEN) and generalized bullous fixed drug eruption (GBFDE) have been reported very rarely in association with the use of NSAIDs, including mefenamic acid. Patients appear to be at highest risk for these events early in the course of therapy, the onset of the event occurring in the majority of cases within the first month of treatment. Mefenamic acid should be discontinued at the first appearance of a skin rash, mucosal lesions, or any other sign of hypersensitivity.

Laboratory test: A false-positive reaction for urinary bile, using the diazo table test, may result following mefenamic acid administration. If biliuria is suspected, other diagnostic procedures, such as the Harrison spot test, should be performed.

Renal Effects: In rare cases, NSAIDs, including mefenamic acid, may cause interstitial nephritis, glomerulitis, papillary necrosis and the nephrotic syndrome. NSAIDs inhibit the synthesis of renal prostaglandin which plays a supportive role in maintenance of renal perfusion in patients whose renal blood flow and blood volume are decreased. In these patients, administration of an NSAID may precipitate overt renal decompensation, which is typically followed by recovery to pretreatment state upon discontinuation of NSAID therapy. Patients at greatest risk of such a reaction are those with congestive heart failure, liver cirrhosis, nephrotic syndrome, overt renal disease, and the elderly. Such patients should be carefully monitored while receiving NSAID therapy.

Discontinuation of NSAID therapy is typically followed by recovery to the pretreatment state. Since mefenamic acid metabolites are eliminated primarily by the kidneys, the drug should not be administered to patients with significantly impaired renal function.

Hematologic Effects: Mefenamic acid can inhibit platelet aggregation and may prolong prothrombin time in patients on warfarin therapy.

Hepatic Effects: Borderline elevations of one or more liver function tests may occur in some patients receiving mefenamic acid therapy. These elevations may progress, may remain essentially unchanged, or may be transient with continued therapy. A patient with symptoms and/or signs suggesting liver dysfunction, or in whom an abnormal liver test has occurred, should be evaluated for evidence of the development of a more severe hepatic reaction while on therapy with mefenamic acid. If abnormal liver tests persist or worsen, if clinical signs and symptoms consistent with liver disease develop, or if systemic manifestations occur, mefenamic acid should be discontinued.

Use of Oral Anticoagulants: The concomitant use of NSAIDs, including mefenamic acid, with oral anticoagulants increases the risk of GI and non-GI bleeding and should be given with caution. Oral anticoagulants include warfarin/coumarin-type and novel oral anticoagulants (e.g., apixaban, dabigatran, rivaroxaban). Anticoagulation/INR should be monitored in patients taking a warfarin/coumarin-type anticoagulant.

INTERACTIONS WITH OTHER MEDICAMENTS

Concurrent therapy with other plasma protein binding drugs may necessitate a modification in dosage.

Anticoagulants: NSAIDs may enhance the effect of anticoagulants such as warfarin. Concurrent administration of mefenamic acid with oral anticoagulant drugs requires careful prothrombin time monitoring. It is considered unsafe to take NSAIDs in combination with warfarin or heparin unless under direct medical supervision.

Lithium: A reduction in renal lithium clearance and elevation of plasma lithium levels. Patients should be observed carefully for signs of lithium toxicity. The following interactions have been reported with NSAIDs but have not necessarily been associated with mefenamic acid.

Other analgesics including cyclooxygenase-2 selective inhibitors: Avoid concomitant use of two or more NSAIDs (including aspirin) as this may increase the risk of adverse effects

Antidepressants: Selective serotonin reuptake inhibitors (SSRIs): increased risk of gastrointestinal bleeding.

Antihypertensive and diuretics: A reduction in antihypertensive and diuretic effect has been observed. Diuretics can increase the nephrotoxicity of NSAIDs.

ACE inhibitors and angiotensin-II-receptor antagonists: A reduction in antihypertensive effect and an increased risk of renal impairment especially in elderly patients. Patients should be adequately hydrated, and the renal function assessed in the beginning and during concomitant therapy.

Aminoglycosides: Reduction in renal function in susceptible individuals decreased elimination of aminoglycoside and increased plasma concentrations.

Anti-platelet agents: Increased risk of gastrointestinal ulceration or bleeding.

Acetylsalicylic Acid: Experimental data implies that mefenamic acid interferes with the anti-platelet effect of low dose aspirin when given concomitantly, and thus may interfere with aspirin's prophylactic treatment of cardiovascular disease. However, the limitations of this experimental data and the uncertainties regarding extrapolation of ex vivo data to the clinical situation imply that no firm conclusions can be made for regular mefenamic acid use.

Cardiac glycosides: NSAIDs may exacerbate cardiac failure, reduce GFR and increase plasma cardiac glycoside levels.

Cyclosporin: The risk of nephrotoxicity of cyclosporin may be increased with NSAIDs.

Corticosteroids: Concomitant use may increase the risk of gastrointestinal ulceration or bleeding.

Oral hypoglycaemic agents: Inhibition of metabolism of sulfonylurea drugs, prolonged half-life and increased risk of hypoglycaemia.

Methotrexate: Elimination of the drug can be reduced, resulting in increased plasma levels.

Mifepristone: NSAIDs should not be taken for 8-12 days after mifepristone administration, NSAIDs can reduce the effects of mifepristone.

Probenecid: Reduction in metabolism and elimination of NSAIDs and metabolites.

Quinolone antibiotics: Animal data indicated that NSAIDs can increase the risk of convulsions associated with quinolone antibiotics. Patients taking NSAIDs and quinolones may have an increased risk of developing convulsions.

Tacrolimus: Possible increased risk of nephrotoxicity when NSAIDs are given with tacrolimus.

Zidovudine: Increased risk of haematological toxicity when NSAIDs are given with zidovudine. There is evidence of an increased risk of hemarthroses and haematoma in HIV(+) haemophiliacs receiving concurrent treatment with zidovudine and ibuprofen.

FERTILITY, PREGNANCY AND LACTATION

Fertility: Based on the mechanism of action, the use of NSAIDs may delay or prevent rupture of ovarian follicles, which has been associated with reversible infertility in some women. In women who have difficulties conceiving or who are undergoing investigation of infertility, withdrawal of NSAIDs, including mefenamic acid should be considered.

Pregnancy: Since there are no adequate and well-controlled studies in pregnant women, this drug should be used only if the potential benefits to the mother justify the possible risks to the foetus. It is not known if mefenamic acid or its metabolites cross the placenta. However, because of the effects of drugs in this class (i.e., inhibitors of prostaglandin synthesis) on the fetal CV system (e.g., premature closure of the ductus arteriosus), the use of mefenamic acid in pregnant women is not recommended and should be avoided during the third trimester of pregnancy. Mefenamic acid inhibits prostaglandin synthesis which may result in prolongation of pregnancy and interference with labour when administered late in the pregnancy. Women on mefenamic acid therapy should consult their physician if they decide to become pregnant.

Inhibition of prostaglandin synthesis might adversely affect pregnancy. Data from epidemiological studies suggest an increased risk of spontaneous abortion after use of prostaglandin synthesis inhibitors in early pregnancy. In animals, administration of prostaglandin synthesis inhibitors has been shown to result in increased pre- and post-implantation loss.

If used during second or third trimester of pregnancy, NSAIDs may cause fetal renal dysfunction which may result in reduction of amniotic fluid volume or oligohydramnios in severe cases. Such effects may occur

shortly after treatment initiation and are usually reversible upon discontinuation. Pregnant women on mefenamic acid should be closely monitored for amniotic fluid volume.

Lactation: Trace amounts of mefenamic acid may be present in breast milk and transmitted to the nursing infant. Therefore, mefenamic acid should not be taken by nursing mothers.

Effects on ability to drive and use machines

The effect of mefenamic acid on the ability to drive or use machinery has not been systematically evaluated.

ADVERSE REACTIONS

Blood and lymphatic system disorders: Agranulocytosis, aplastic anaemia, autoimmune haemolytic anaemia*, bone marrow hypoplasia, decreased haematocrit, eosinophilia, leukopenia, pancytopenia, thrombocytopenic purpura, platelet aggregation inhibition.

Immune system disorders: Anaphylaxis

Metabolism and nutrition disorders: Glucose intolerance in diabetic patients, hyponatremia, fluid retention

Psychiatric disorders: Nervousness

Nervous system disorders: Aseptic meningitis, blurred vision, convulsions, dizziness, drowsiness, headache, and insomnia

Eye disorders: Eye irritation, reversible loss of colour vision

Ear and labyrinth disorders: Ear pain

Cardiac disorders: Palpitation

Vascular disorders: Hypotension, hypertension

Respiratory, thoracic, and mediastinal disorders: Asthma, dyspnoea

Gastrointestinal disorders: Gastrointestinal inflammation, gastrointestinal haemorrhage, gastrointestinal ulcer, gastrointestinal perforation.

The most frequently reported side effects associated with mefenamic acid involve the gastrointestinal tract. Diarrhoea appears to be the most common side effect and is usually dose-related. It generally subsides on dosage reduction, and rapidly disappears on termination of therapy. Some patients may not be able to continue therapy.

The following are the most common gastrointestinal side effects: Abdominal pain, diarrhoea, and nausea with or without vomiting.

Less frequently reported gastrointestinal/hepatobiliary side effects: Anorexia, cholestatic jaundice, colitis, constipation, enterocolitis, flatulence, gastric ulceration with and without haemorrhage, mild hepatic toxicity, hepatitis, hepatorenal syndrome, pyrosis, pancreatitis and steatorrhea.

Skin and subcutaneous tissue disorders: Angioedema, oedema of the larynx, erythema multiforme, facial oedema, Lyell's syndrome (toxic epidermal necrolysis),

perspiration, pruritus, rash, Stevens-Johnson syndrome, urticaria, dermatitis exfoliative, and generalized bullous fixed drug eruption (GBFDE).

Renal and urinary disorders: Dysuria, haematuria, renal failure including papillary necrosis and tubulointerstitial nephritis, glomerulonephritis, nephrotic syndrome.

General disorders and administration site conditions: Oedema

Investigations: Urobilinogen urine (false-positive), liver function test abnormal

Paediatric patients

General disorders and administration site conditions: Hypothermia

*Reports are associated with ≥12 months of mefenamic acid therapy and the anaemia is reversible with discontinuation of therapy.

SYMPTOMS AND TREATMENT FOR OVERDOSAGE

Seizures, acute renal failure, coma, confusional state, vertigo, and hallucination have been reported with mefenamic acid overdoses. Overdose has led to fatalities.

For oral formulations:

Patients should be managed by symptomatic and supportive care following an NSAID overdose. There are no specific antidotes. Following acute overdosage, emesis, and/or gastric lavage, and/or activated charcoal may be considered dependent upon amount ingested and time since ingestion. Vital functions should be monitored and supported. Haemodialysis is of little value since mefenamic acid and its metabolites are firmly bound to plasma proteins.

PACKING SIZES

100 Strips of 10 Capsules

STORAGE CONDITIONS

Store below 30°C; away from direct light.

SHELF LIFE: Product should be not used beyond the expiry date imprinted on the product packaging.

REGISTRATION NUMBER: MAL19860139AZ

For further information, please consult your doctor or your pharmacist.

Revision Number: 08

Revision Date : 28 October 2024

Lt-106.06 Product Registration Holder and Manufacturer

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